**Wrocław University of Science and Technology**
Faculty of Management

Descriptive analytics

# First task set report

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# 1 Task 1.1

## 1.1 Description

The aim of the first task was to empirically verify the relationship between independent standard normal variables and the chi-squared distribution. Specifically, the task demonstrates that the sum of the squares of the $k$ independent standard normal random variables follows a chi-squared distribution with $k$ degrees of freedom ($Z \sim \chi^2(k)$). In this case, we analyzed the sum of three squared samples ($k = 3$) to observe how closely the empirical data align with the theoretical mathematical models.

## 1.2 Subplot

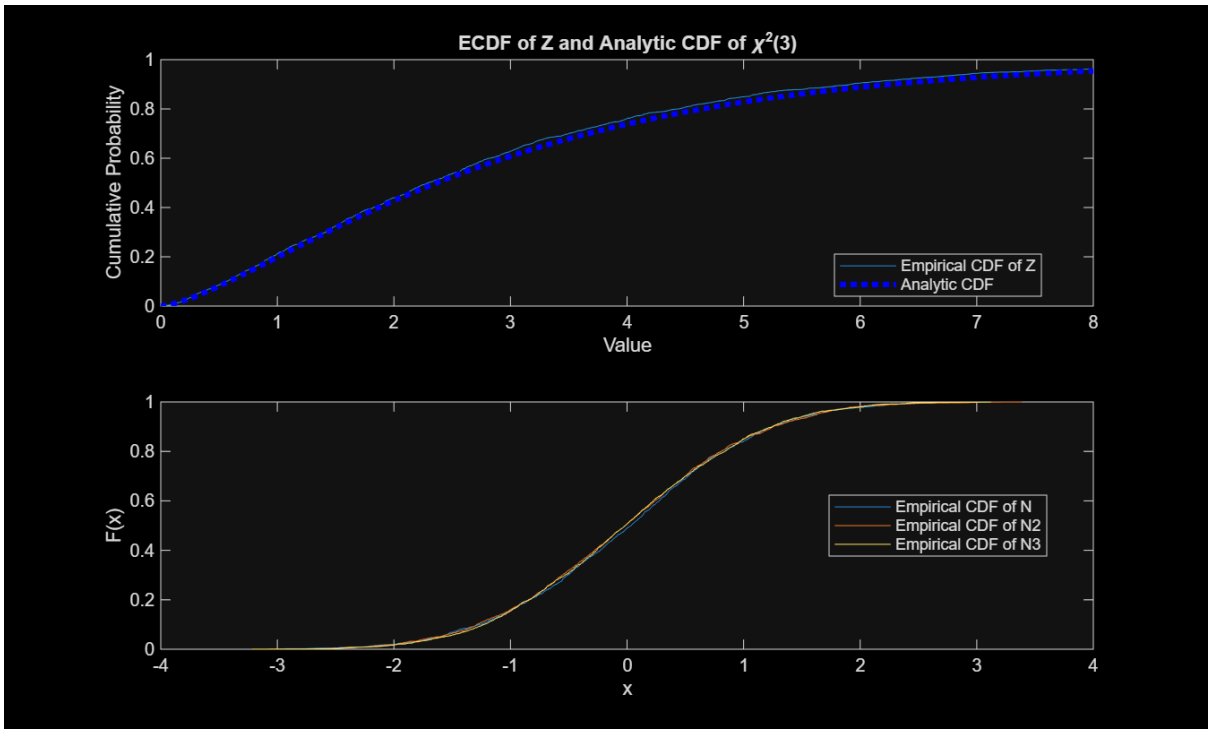


Figure 1: ECDF of Z and Analytic CDF of $\chi^2(3)$

### 1.3 Analytic statistics comparison

The table below summarizes the comparison between the empirical results (mean, median, variance, and skewness) of our generated sample $Z$ and the analytic properties of the $\chi^2(3)$ distribution.

| | Zetki | Chi |
|---|---|---|
| Mean | 2.8542 | 3 |
| Median | 2.2951 | 2.366 |
| Variance | 5.2517 | 6 |
| Skewness | 1.5568 | 1.633 |

Figure 2: Table with basic analytic statistics of variable Z and $\chi^2(3)$

By comparing the ECDF of the transformed variable $Z$ with the theoretical curve $\chi^2(3)$, we can conclude whether the simulation successfully approximates the expected distribution. The alignment in the upper plot, coupled with the convergence of the mean and variance in the table above, provides numerical and visual evidence of the fundamental link between Normal and Chi-Squared distributions.

## 2 Task 1.2

### 2.1 Description

The aim of the second task was to explore non-parametric density estimation and identify the underlying distribution of provided data samples. By applying Kernel Density Estimation (KDE) with varying parameters, our objective was to observe how the choice of kernel function and bandwidth ($h$) affects the smoothness and accuracy of the estimated probability density function. Finally, the task required identifying which sample originates from a normal distribution using visual cues and verifying the hypothesis via Quantile-Quantile (Q-Q) plots.

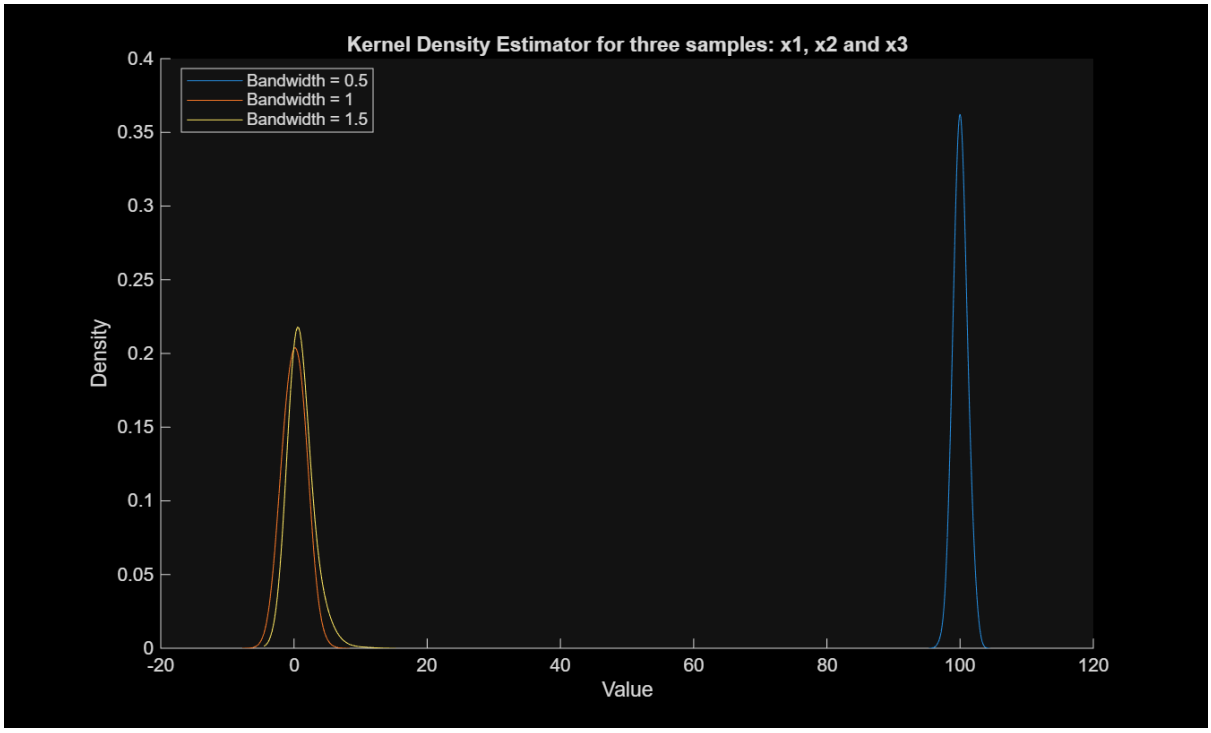


Figure 3: KDE for three samples - X1, X2 and X3

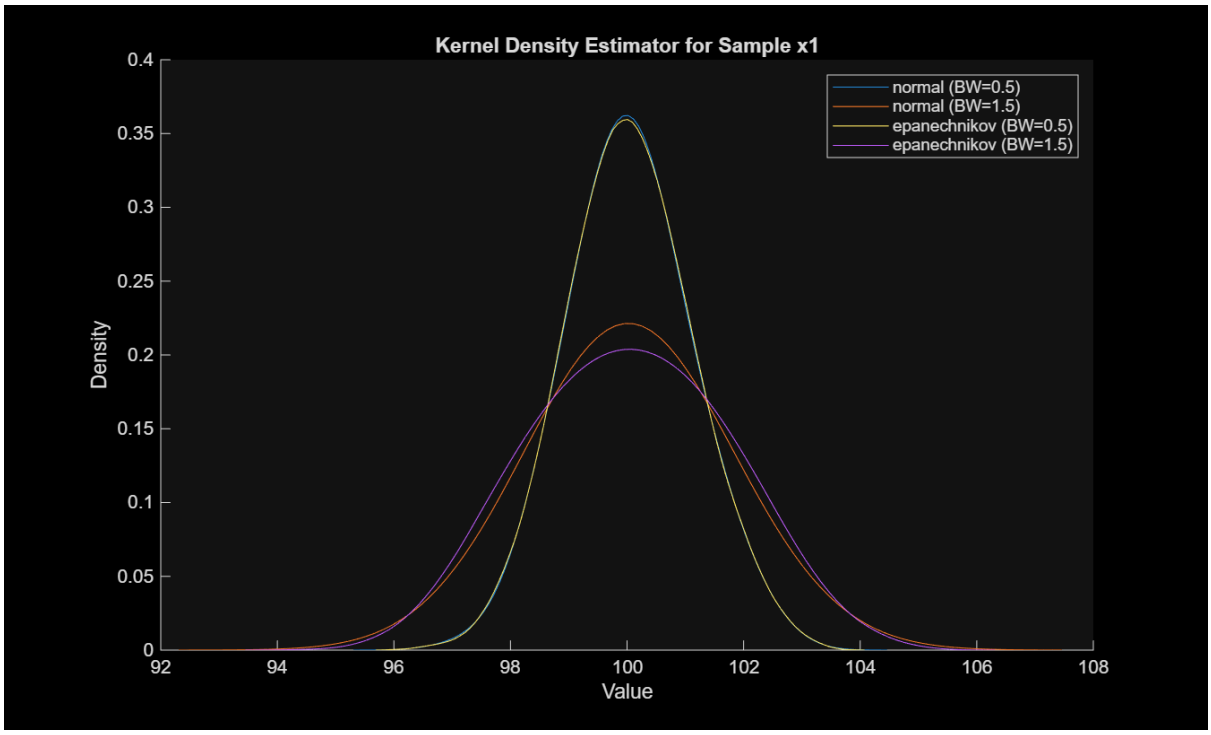


Figure 4: KDE for sample X1

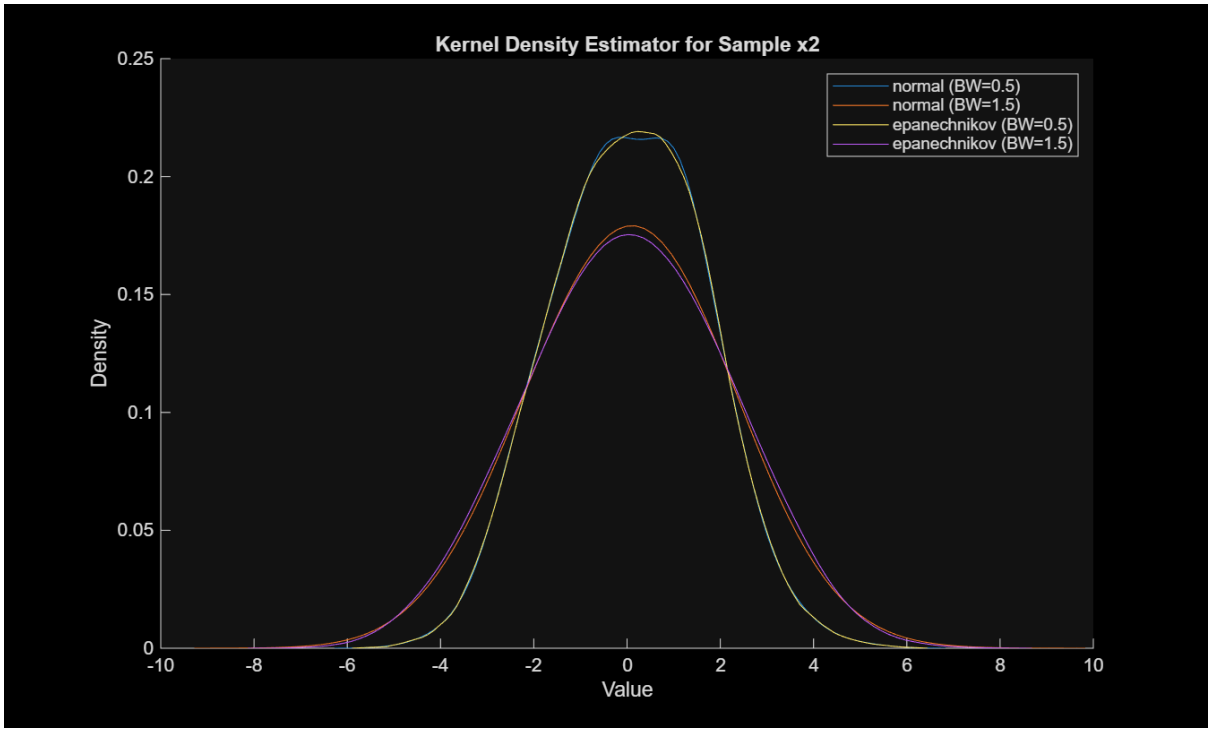


Figure 5: KDE for sample X2

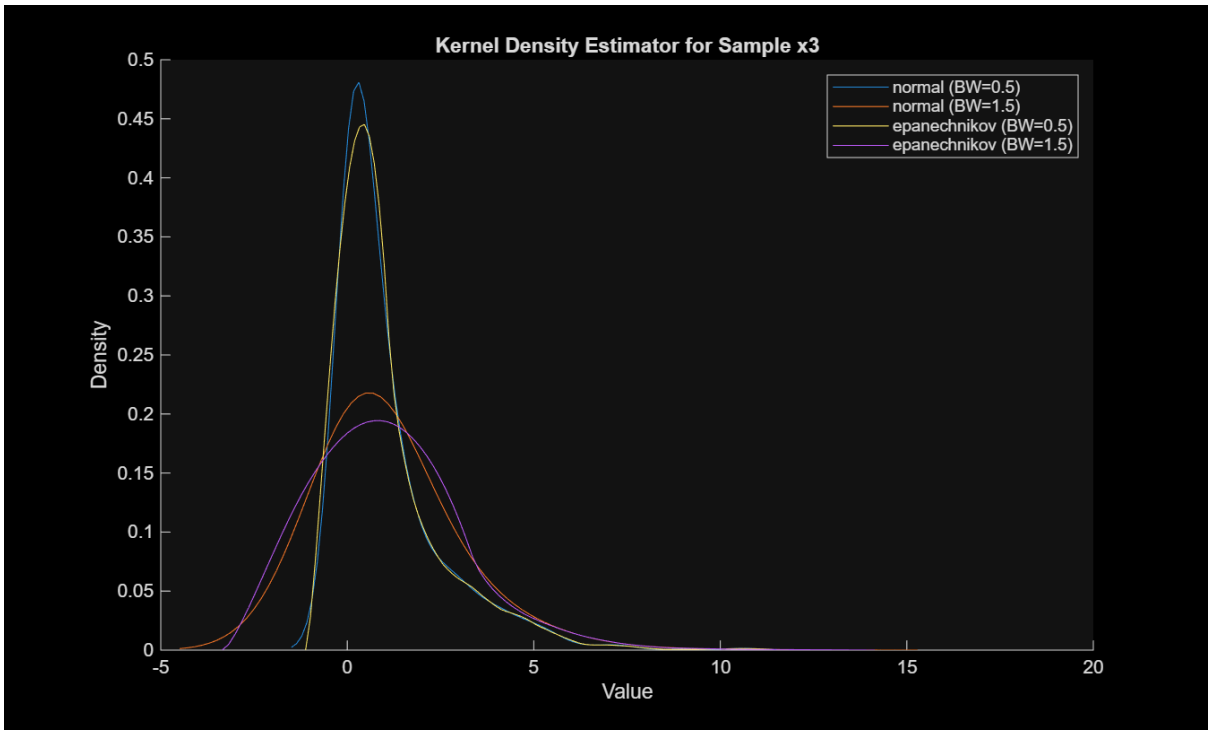


Figure 6: KDE for sample X3

In my opinion, the first sample actually comes from a natural distribution. That's because its plots are really similar to the Gaussian function - they intensively resemble its very original "bell shape". Via Q-Q plots we can definitely verify this assumption.

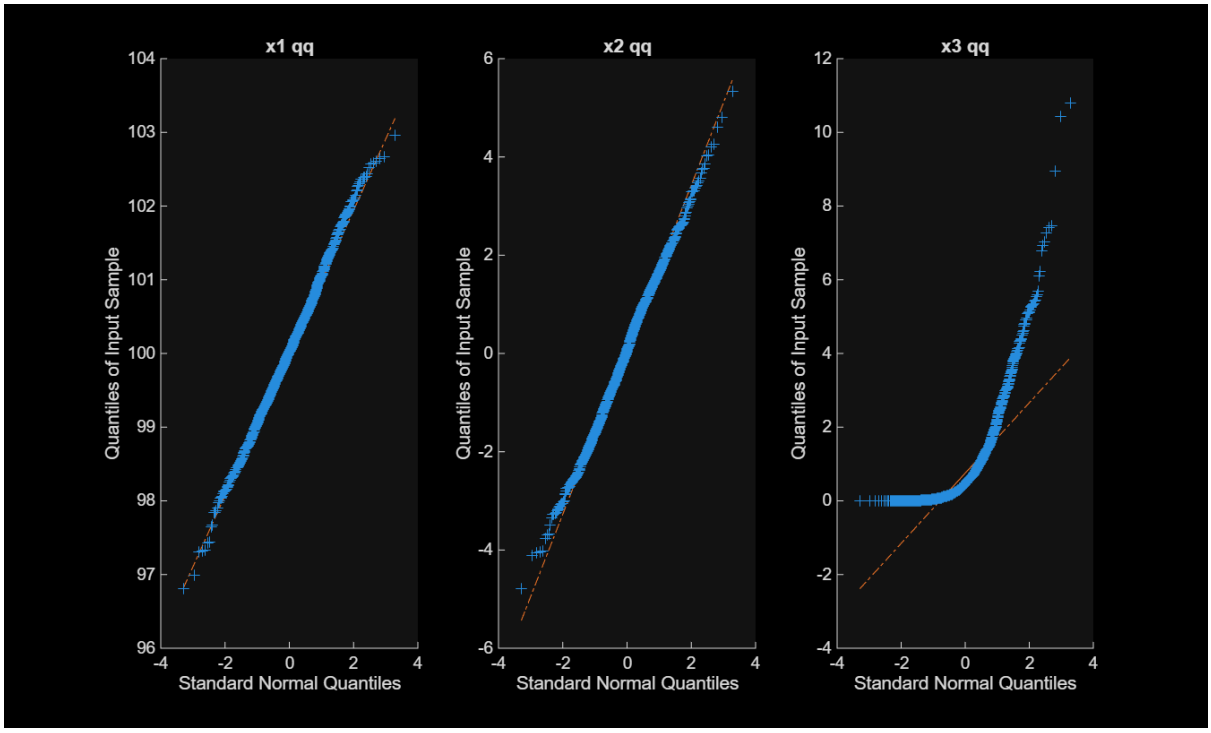


Figure 7: Q-Q Plot of three samples

## 2.2 Conclusion

This task demonstrated that KDE is a powerful tool for visualizing the distribution of data without assuming a specific parametric form. However, the accuracy of the visualization is highly dependent on the bandwidth parameter. The subsequent Q-Q plot analysis provided a more rigorous statistical confirmation of normality than simple visual inspection of the density curve alone.

# 3 Task 1.3

## 3.1 Description

The purpose of the third task was to perform a comparative statistical analysis of data samples to identify the underlying distributions and quantify the relationships between the variables. This involved applying formal normality tests to distinguish between normal and non-normal data, identifying the distribution type of non-normal samples through visual estimation, and evaluating linear versus monotonic associations using Pearson and Spearman correlation coefficients.

## 3.2 Normality test

The Lilliefors test is a modification of the Kolmogorov–Smirnov test used to check whether a sample comes from a normally distributed population when the mean and variance are estimated from the data. Unlike the standard KS test, it adjusts the critical values to account for this estimation. It is commonly used as a goodness-of-fit test for normality in practical data analysis. The results are as follows:

```
testx = lillietest(x) % 1 - nienormalne
testy1 = lillietest(y1) % 0 - normalne
```

Listing 1: Lilliefors test result

The Lilliefors test results have shown that the Y1 sample is of normal descent contrary to the X sample. Let us verify this result by plotting the samples.

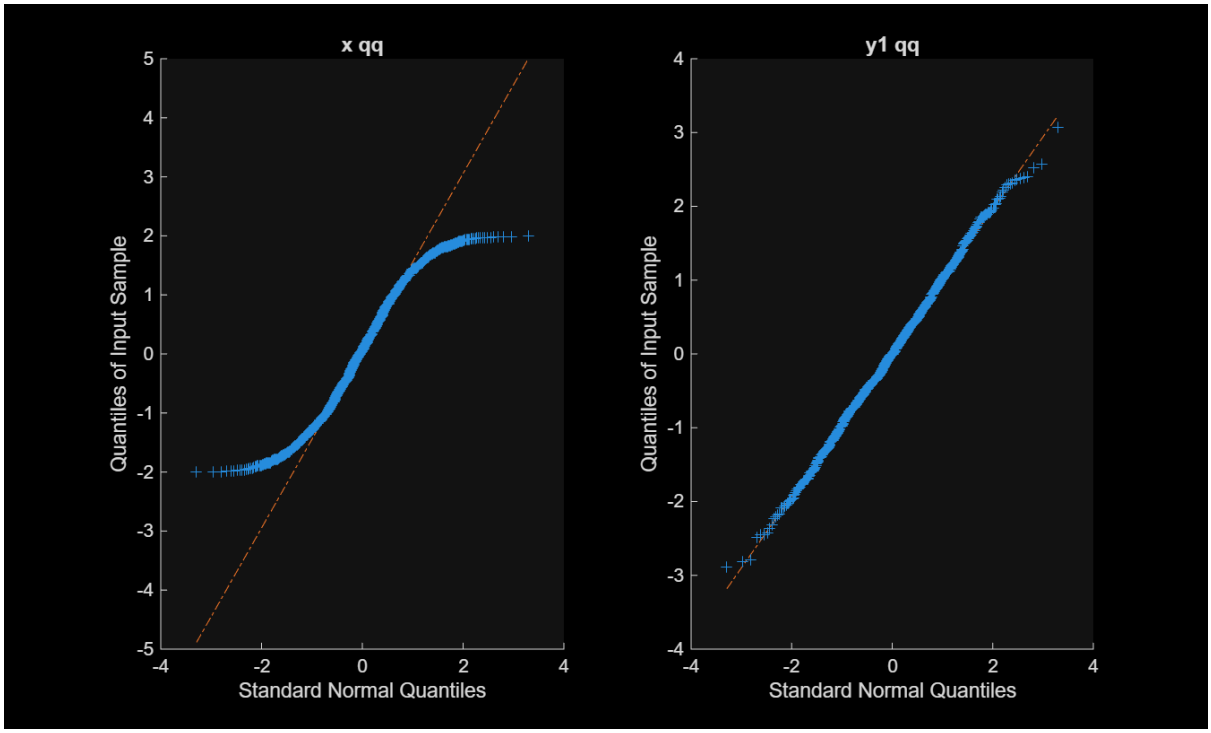


Figure 8: Q-Q Plot of two samples

The Q-Q plot confirms the results of the earlier Lilliefors test. With this being said, we can officially acknowledge the Y1 sample as a descendant of the Gaussian function.

### 3.3 Plot of sample X

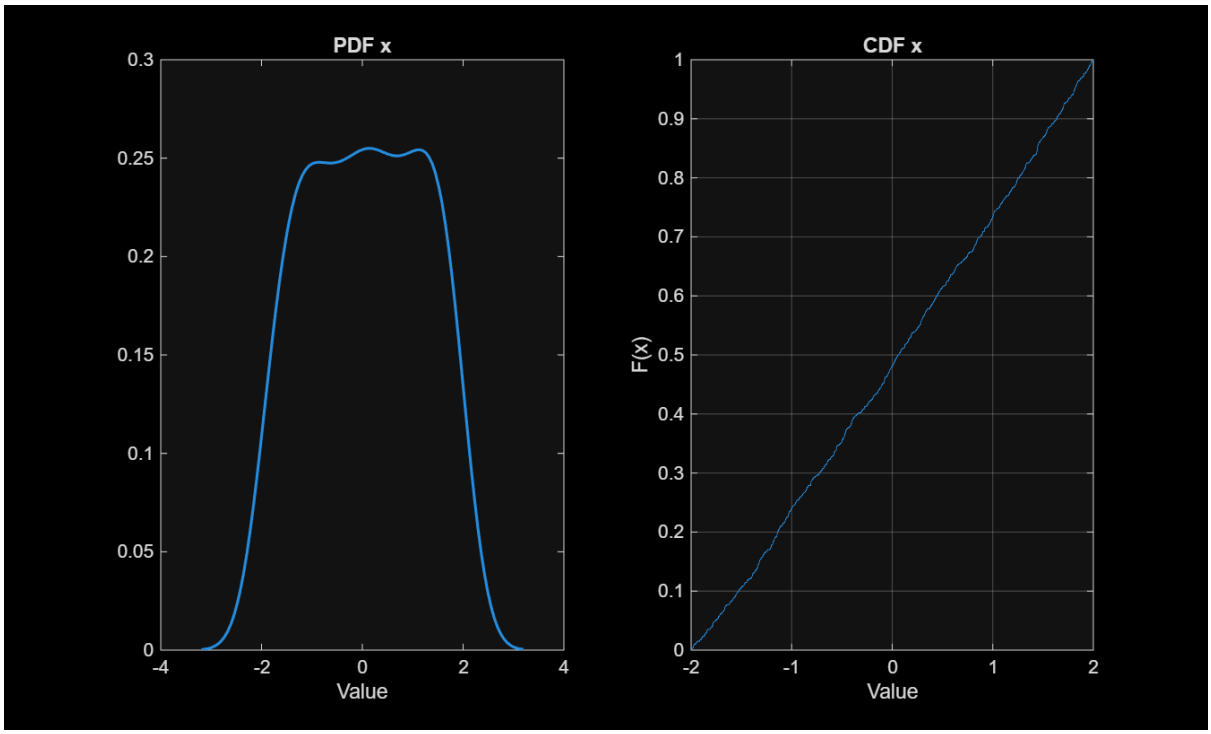


Figure 9: CDF and PDF estimator of a not normal vector

### 3.4 Correlation between pairs of samples

```
pearson_1 = corr(x', y1', "Type", "Pearson")
spearman_1 = corr(x', y1', "Type", "Spearman")

% Pearson_1 = 0.0187 ; Spearman_1 = 0.0249
% X and Y1 are little to none positevly correlated

pearson_2 = corr(x', y2', "Type", "Pearson")
spearman_2 = corr(x', y2', "Type", "Spearman")

% Pearson_2 = -0.9110 ; Spearman_2 = -0.9685
% X and Y2 are definitely negatively correlated

pearson_3 = corr(x', y3', "Type", "Pearson")
spearman_3 = corr(x', y3', "Type", "Spearman")

% Pearson_3 = 0.8709 ; Spearman_3 = 1
% X and Y3 are definitely positively correlated
```

Listing 2: Correlation test result

# 4 Additional Task 1.4

## 4.1 Description

The purpose of this task was to perform a comparative statistical analysis of stock market data for the year 2022. By calculating daily percentage returns, the study focused on characterizing the volatility, distribution, and correlation of AVGO and STT. Key objectives included testing whether average returns differ significantly from zero, identifying appropriate correlation measures for financial assets, and evaluating the normality of financial returns using specialized statistical tests.

## 4.2 Returns of both indices and basic analytic statistics

```
Piec zwrotow z inwestycji dla AVGO
   -0.3146
    1.1460
   -4.1614
   -0.9285
   -2.8069

Piec zwrotow z inwestycji dla STT
    1.5601
    4.0974
   -1.1901
    1.8218
    2.7685
```

Listing 3: Returns of both indices

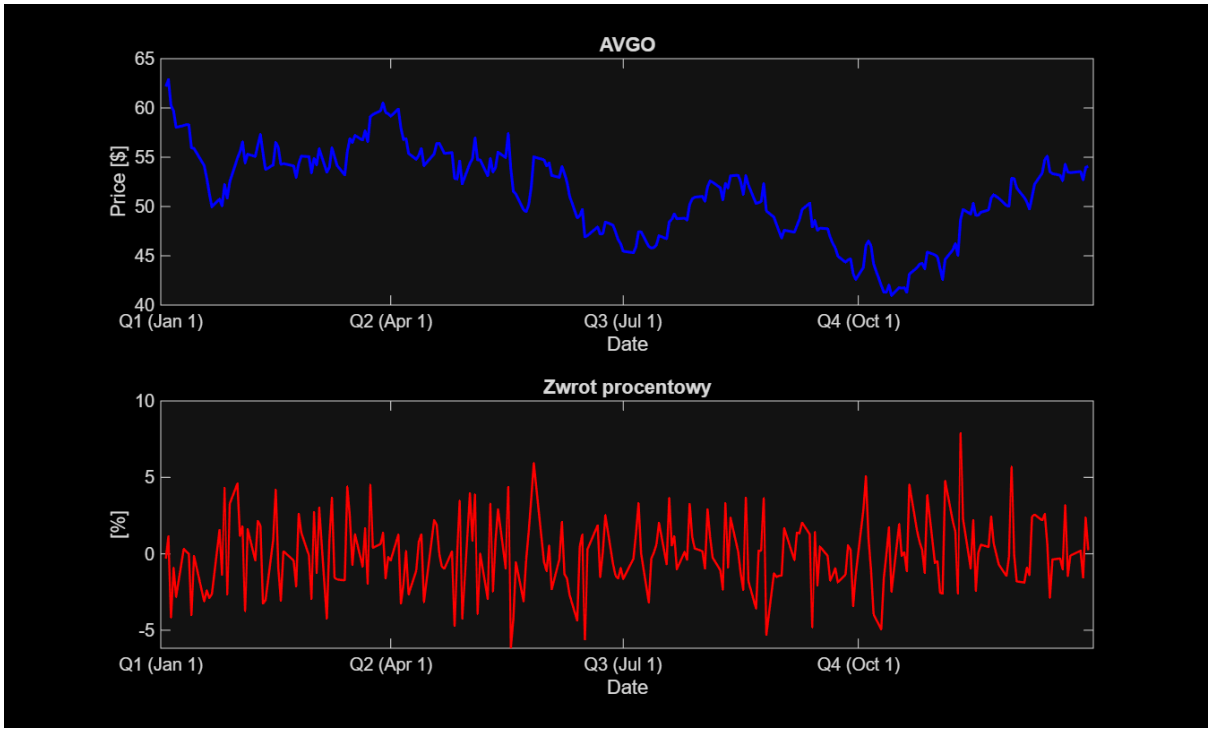


Figure 10: AVGO return rates

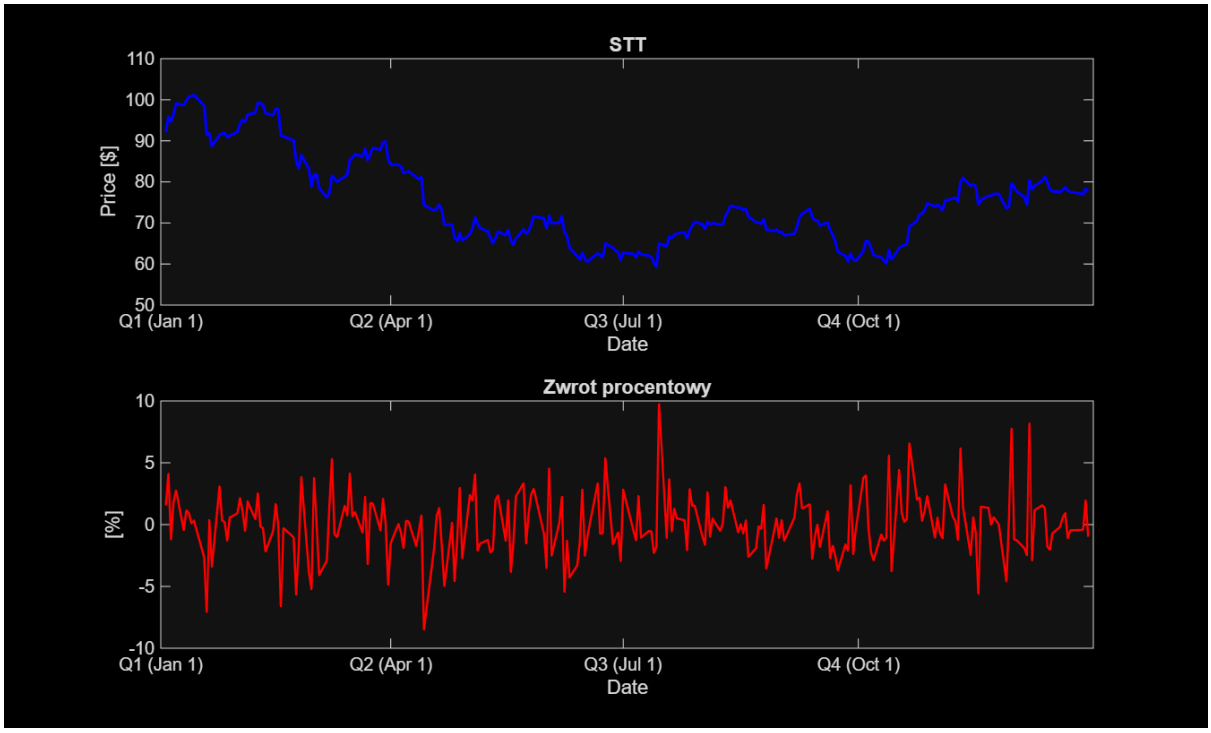


Figure 11: STT return rates

| | tabela_avgo | tabela_stt |
|---|---|---|
| Mean | -0.029262 | -0.029983 |
| Standard Deviation | 2.3496 | 2.5443 |
| Median | -0.097808 | -0.17846 |
| MAD | 1.8432 | 1.9142 |
| Skewness | 0.18353 | 0.24866 |
| IQR | 2.9364 | 3.0346 |

Figure 12: Comparison between both indices' analytic statistics

### 4.3 Returns check using T test

```
alfa = 0.01;

[h_avgo, p_avgo] = ttest(zwroty_avgo, 0, 'Alpha', alfa);
disp(['P-value to ', num2str(p_avgo)]);
disp(['Jaka decyzja = ', num2str(h_avgo)]);

% P-value = 0.84375 ; Decyzja = 0

[h_stt, p_stt] = ttest(zwroty_stt, 0, 'Alpha', alfa);
disp(['P-value to ', num2str(p_stt)]);
disp(['Jaka decyzja = ', num2str(h_stt)]);
```

```
% P-value = 0.85205 ; Decyzja = 0
```

Listing 4: Testing the correlation between two indices

## 4.4 Correlation between two indices

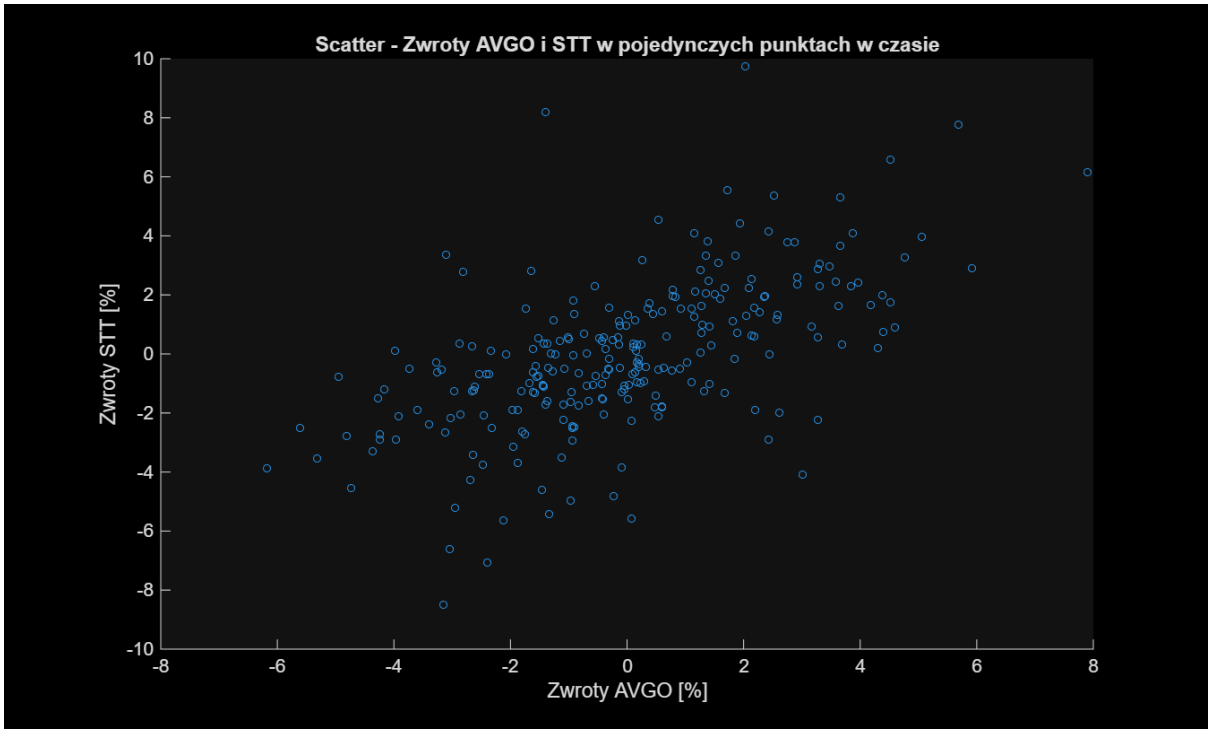


Figure 13: Correlation between two indices

The returns of both indices are not definitely but strongly correlated, with Pearson and Spearman tests oscillating around 0.6 values. This means that both companies are experiencing growth in the same time. When it comes to choosing the valid test, I would choose Spearman. That is because it is not easily manipulated by outliers in datasets, which is critically important, especially in finances, to predict future values of shares and returns.

## 4.5 Distribution and normality tests

```
[h_ks, p_ks] = kstest2(zwroty_avgo, zwroty_stt);

disp(['P-value to ', num2str(p_ks)]);
disp(['Jaka decyzja = ', num2str(h_ks)]);

% P-value = 0.88109 ; Decyzja = 0
```

Listing 5: Testing whether the two indices have the same distribution

The achieved P-value is much larger than the alpha - that is why there is no reason to reject null hypothesis, meaning that both indices have the same distribution.
The Kolmogorov-Smirnov test is not an adequate choice in this situation, because it is mainly used for cases where parameters are known from the start. Here, because we calculate parameters inside the code, a Jarque-Bera test would be better. This is a test widely used and recommended in the financial sector.

```
[h_jb_avgo, p_jb_avgo] = jbtest(zwroty_avgo);
[h_jb_stt, p_jb_stt]   = jbtest(zwroty_stt);

disp(['P-value AVGO to ', num2str(p_jb_avgo)]);
disp(['P-value STT to ', num2str(p_jb_stt)]);

% P-value AVGO = 0.43707 ; P-value STT = 0.0013374
```

Listing 6: Checking normality of both indices

In contrast to AVGO, the P-value of STT is much lower than alpha. That is why in this case we can reject the null hypothesis and assume that the distribution of STT is not normal.

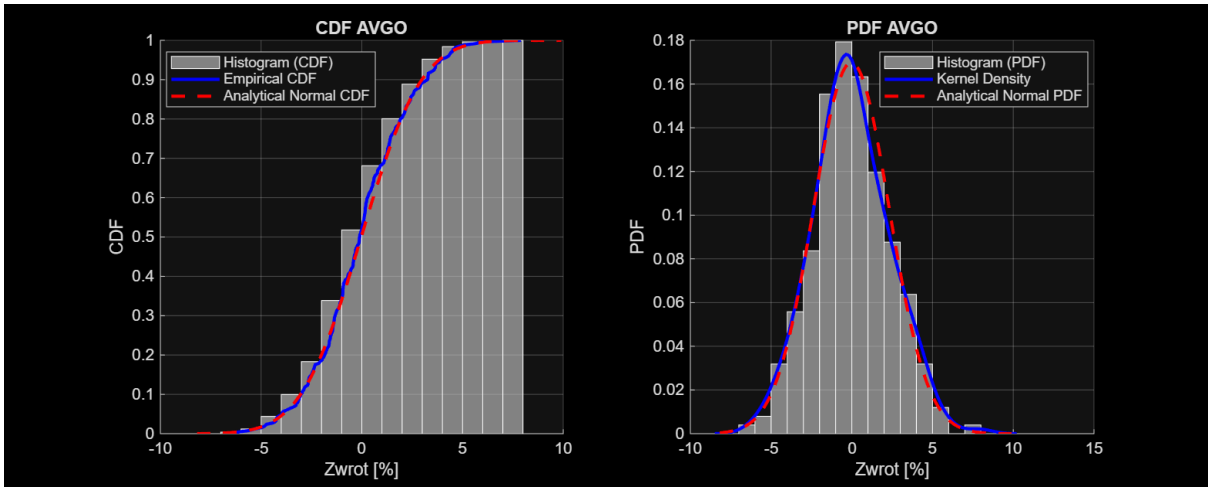


Figure 14: ECDF and Kernel estimated density function for the more normal distributed indice